

Week 11 – Skill

```
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

#define MAX_HISTORY 10
#define MAX_CMD 100

int main() {
    char history[MAX_HISTORY][MAX_CMD];
    char command[MAX_CMD];
    int count = 0;

    printf("Simple Command History\n");
    printf("Type 'history' to display commands\n");
    printf("Type 'exit' to quit\n\n");

    while (1) {
        printf("shell> ");

        if (fgets(command, sizeof(command), stdin) == NULL)
            break;

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
            break;

        if (strcmp(command, "history") == 0) {
            printf("\nCommand History:\n");

            for (int i = 0; i < count; i++)
                printf("%d %s\n", i + 1, history[i]);

            continue;
        }
    }
}
```

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```
    if (fgetc(command, sizeof(command), stdin) == NULL)
        break;

    command[strcspn(command, "\n")] = '\0';

    if (strcmp(command, "exit") == 0)
        break;

    if (strcmp(command, "history") == 0) {
        printf("\nCommand History:\n");

        for (int i = 0; i < count; i++)
            printf("%d %s\n", i + 1, history[i]);

        continue;
    }

    if (strlen(command) == 0)
        continue;

    if (count < MAX_HISTORY) {
        strcpy(history[count], command);
        count++;
    } else {
        for (int i = 0; i < MAX_HISTORY - 1; i++)
            strcpy(history[i], history[i + 1]);

        strcpy(history[MAX_HISTORY - 1], command);
    }
}

printf("Exiting shell...\n");
return 0;
}
```

```
samanth-ubuntu@samanth-ubuntu:~/skill11$ nano skill-11.c
samanth-ubuntu@samanth-ubuntu:~/skill11$ gcc skill-11.c -o skill-11
samanth-ubuntu@samanth-ubuntu:~/skill11$ ls -l skill-11
-rwxrwxr-x 1 samanth-ubuntu samanth-ubuntu 70808 Oct  9 09:36 skill-11
samanth-ubuntu@samanth-ubuntu:~/skill11$ ./skill-11
Simple Command History
Type 'history' to display commands
Type 'exit' to quit

shell> ls
shell> pwd
shell> date
shell> history

Command History:
1 ls
2 pwd
3 date
shell> █
```